Additional file 1: Supplementary methods and results

Companion to the Reproductive Health manuscript draft

25 October 2026

Additional arithmetic definitions

The same-population benchmark is $B_T = \sum_a P_{C,aT}(r_{C,aT} - r_{R,aT})/10^5$. It uses China's actual age distribution rather than standard weights and is not preventable burden. Age-specific contributions to the standardized difference are $w_a(r_{C,at} - r_{R,at})$. The seven published unnormalized standard weights, in ascending age order, are 0.082891, 0.078012, 0.075914, 0.073217, 0.068281, 0.061474 and 0.055113; they sum to 0.494902 before renormalization. References and outcome definitions are given in the main manuscript.

1 Supplementary methods and results

1.1 Decomposition of case-count changes

We expressed the case count as $F(N, \boldsymbol{f}, \boldsymbol{p}) = N \sum_a f_a p_a$, where N is the total female population aged 15–49, f_a is its age distribution, and p_a is age-specific prevalence as a proportion. For each factor, we averaged its marginal contribution across all six possible replacement orders between the starting and ending states. These Shapley contributions sum exactly to the observed modeled case-count change. We examined 1990–2023 and 2010–2023 for both causes in all 19 countries. Population size here refers specifically to females aged 15–49; age composition refers to redistribution within this range. The age-specific prevalence component includes epidemiological change, ascertainment and modeling changes and cannot be interpreted as a causal risk-factor contribution.

1.2 Trends, sensitivities and forecast validation

We fitted continuous piecewise linear models to log standardized rates, searching integer-year knot locations with zero to two knots and at least five observations per segment, counting shared boundary years in each segment. Model selection used the Bayesian information criterion, including a penalty for knot locations. Segment-specific annual percentage change was $100[\exp(\beta) - 1]$; average annual percentage change (AAPC) was obtained from duration-weighted log slopes. These fits summarize modeled trajectories and are not outputs of the proprietary Joinpoint program. AR(1) generalized least-squares refits retained the OLS-selected

knots; allowing up to three knots and restricting the period to 2010–2023 were additional sensitivities. Trend confidence intervals are conditional on selected knots and fixed GBD point estimates.

Reference sensitivities used age-specific country population shares fixed at 1990, excluded one reference country at a time, and restricted ages to 20–49 with renormalized weights. Following inspection of the terminal-year endometriosis increase, we added analyses ending in 2021 and 2022; these endpoint analyses were not prespecified. To evaluate numerical sensitivity to published weight rounding, we evaluated all 2^7 combinations of perturbing unnormalized weights by $\pm 0.5 \times 10^{-6}$. The resulting deterministic envelopes are not uncertainty intervals.

An exploratory forecasting module evaluated each Chinese age-specific prevalence series on the logit scale. Candidates were a constant-last-rate baseline, ARIMA with $p, q \in \{0, 1, 2\}$ and $d \in \{0, 1\}$ selected by training-set corrected AIC, and additive damped-trend exponential smoothing. ARIMA candidates failing convergence were excluded. Rolling training windows contained at least 20 years and assessed horizons 1, 3 and 5; a separate last-10-year holdout assessed horizons 1–10. We specified an exploratory retention rule before running this module: at least 10% lower mean absolute error (MAE) than the baseline in every age group in both validation schemes, with complete predictions. Future counts would additionally require compatible population projections; none were generated when validation failed. This module was not a validation against independent observed infertility data.

Table S1: Shapley decomposition of changes in China's modeled prevalent cases, thousands.

Cause	Period	Net change	Population size	Age composition	Prevalence
Endometriosis	1990–2023	-11.3	-8.7	30.8	-33.4
Endometriosis	2010–2023	6.1	-38.4	7.7	36.8
PCOS	1990–2023	783.4	-68.4	-8.9	860.7
PCOS	2010–2023	-129.8	-414.4	-18.2	302.9

Population size is the female population aged 15–49; age composition is its within-range distribution; prevalence is age-specific modeled infertility prevalence. Contributions may exceed the net change and need not have the same sign. Rounding can cause small displayed discrepancies.

Table S2: Restricted-age standardized YLD rates and YLD totals.

		YLDs, thousands		Rate per 100,000			
Cause	Year	China	Reference	China	Reference	*D*	*RR*
Endometriosis	1990	1.0	2.2	0.34	0.39	-0.05	0.872
Endometriosis	2023	1.0	2.6	0.29	0.31	-0.02	0.934
PCOS	1990	5.9	20.5	1.80	3.59	-1.79	0.502
PCOS	2023	10.1	35.5	3.31	4.22	-0.91	0.784

All values are derived point estimates. YLD: years lived with disability.

Table S3: Endpoint sensitivity added after inspection of terminal-year estimates.

Cause	End year	China ASPR	Ref. ASPR	D	RR	$D_T - D_{1990}$
Endometriosis	2021	43.48	51.93	-8.45	0.837	-5.52
Endometriosis	2022	43.13	51.89	-8.76	0.831	-5.83
Endometriosis	2023	53.30	53.51	-0.20	0.996	2.73
PCOS	2021	601.79	745.23	-143.44	0.808	129.47
PCOS	2022	602.09	726.09	-124.00	0.829	148.91
PCOS	2023	605.08	726.45	-121.38	0.833	151.54

The source release remains GBD 2023 in all rows. Ending in 2021 is not substitution of GBD 2021 estimates. Rate units: per 100,000 women.

Table S4: Conditional trend sensitivity for prevalence, 1990–2023.

Location	Cause	Model	AAPC (%)	Conditional 95% CI
China	Endometriosis	OLS, 0–2 knots	-0.86	(-1.00, -0.71)
China	Endometriosis	AR(1), fixed knots	-0.82	(-0.99, -0.65)
China	Endometriosis	OLS, 0–3 knots	-0.86	(-1.00, -0.71)
China	PCOS	OLS, 0–2 knots	1.89	(1.85, 1.93)
China	PCOS	AR(1), fixed knots	1.94	(1.81, 2.08)
China	PCOS	OLS, 0–3 knots	1.84	(1.80, 1.87)
Reference	Endometriosis	OLS, 0–2 knots	-0.74	(-0.78, -0.71)
Reference	Endometriosis	AR(1), fixed knots	-0.74	(-0.78, -0.69)
Reference	Endometriosis	OLS, 0–3 knots	-0.72	(-0.75, -0.68)
Reference	PCOS	OLS, 0–2 knots	0.53	(0.49, 0.58)
Reference	PCOS	AR(1), fixed knots	0.49	(0.42, 0.57)
Reference	PCOS	OLS, 0–3 knots	0.57	(0.53, 0.61)

Intervals condition on selected knots and fixed GBD point estimates. AR(1) retains the knots selected by OLS with up to two knots. These intervals exclude GBD, knot-selection and structural model uncertainty. Full periods, measures, knots and residual-correlation estimates are in the accompanying CSV.

Table S5: Age-specific forecast-validation MAE ratios relative to the naive baseline.

Cause	Model	Rolling: min–max	Last 10 years: min–max	Retained
Endometriosis	ARIMA	0.87–1.30	0.93–1.80	No
Endometriosis	ETS	1.00–1.51	0.86–2.10	No
PCOS	ARIMA	0.19–2.20	0.06–1.38	No
PCOS	ETS	0.20–1.79	0.09–1.28	No

Ranges span seven age groups. Values below 1 favor the candidate. Retention required every age-specific ratio to be below 0.9 in both schemes, with complete predictions. The rule was specified before running this exploratory module, not externally preregistered. Rolling MAE combines horizons 1, 3 and 5; the holdout combines horizons 1–10.

1.3 Acquisition and computation audit

The official query identifiers were:

Burden: `1b13e3376135ab41a4bc5e0ffc60e84f`.

Population: `916826fb285f8d3c3e53117dd2133406`.

Pilot: `0daed0e53f5ad8ebdf0dc00c23ccd110`.

Annual strata: `416daf735b9fc79bec35506514424e73`.

Endpoint strata: `76c5523c52f32caf342843daf2326fa4`.

Raw uncertainty limits remain in the original CSVs. Country and outcome mappings, source hashes, rounding sensitivities and all analytical tables are separate machine-readable files. Nine automated core checks plus the new reference-composition accounting checks cover core accounting identities, the complete data grid, denominator consistency and outcome gates; temporary artificial test fixtures were not used as research data.

The protocol was written before analysis but was not externally preregistered. The endpoint and weight-rounding sensitivities were added following quality inspection. Forecasting remained exploratory and was withheld after failed comparative validation. Primary/secondary stratification was completed using annual Number exports and independent endpoint Number/Rate checks. The symmetric reference-composition and age-by-type analyses were added during manuscript review and were not prespecified. The separately acquired NSFG respondent analysis was completed post hoc and independently numerically checked; these survey estimates do not validate GBD cause attribution. Future population scenarios were not performed. The annual stratum rate reconstruction reduced redundant metrics within the remaining GBD download allowance.

1.4 GATHER reporting map

This map applies GATHER to the secondary analysis; upstream input reconstruction and primary disease estimation belong to the original GBD study. It is not a claim of independent GBD model replication.

Item	Topic	Location or applicability
1	Indicators and populations	Methods: causes, ages, female sex, fixed countries and 1990–2023.
2	Funding	Declarations: author-supplied information remains pending.
3	Data identification and access	Official Results Tool, download date, query identifiers and cited GBD methods.
4	Eligibility and exclusions	Exact outcome filters and completeness gates; no country removals from the main set. Sensitivity exclusions are explicit.

Item	Topic	Location or applicability
5	Source characteristics	GBD model outputs, dimensional mappings and acquisition audit. Original source-study characteristics are documented by GBD.
6	Potential input biases	Discussion: sparse evidence, diagnostic heterogeneity, attribution and modeled denominators.
7	Other inputs	Matching GBD population and published standard weights with provenance.
8	Input data access	CSVs retained in the research archive; source institution IHME and exact queries identified for reproducibility.
9	Conceptual overview	Equations for within-age pooling, standardization, contrasts and decomposition.
10	Analytical detail	Normalization, validation, comparison, decomposition, trend and sensitivity methods and scripts.
11	Candidate evaluation and selection	Segmented-model BIC, training AICc and explicit forecast-retention criterion.
12	Evaluation results	Reference, endpoint, AR(1), rounding and forecast-validation results.
13	Uncertainty methods and scope	Joint draws unavailable; no propagated derived intervals; conditional trend interval scope stated.
14	Source code access	GitHub repository, pinned dependencies and executable scripts.
15	Machine-readable estimates	Complete derived CSV tables accompany the paper.
16	Quantitative uncertainty	Incomplete for derived estimates: marginal source limits retained but cannot substitute for joint uncertainty. Trend intervals are conditional only.
17	Interpretation and prior evidence	Discussion compares prior GBD studies. Release-to-release differences were not estimated.
18	Limitations	Discussion: endpoint, reference, denominator, source, rounding and inference limitations.

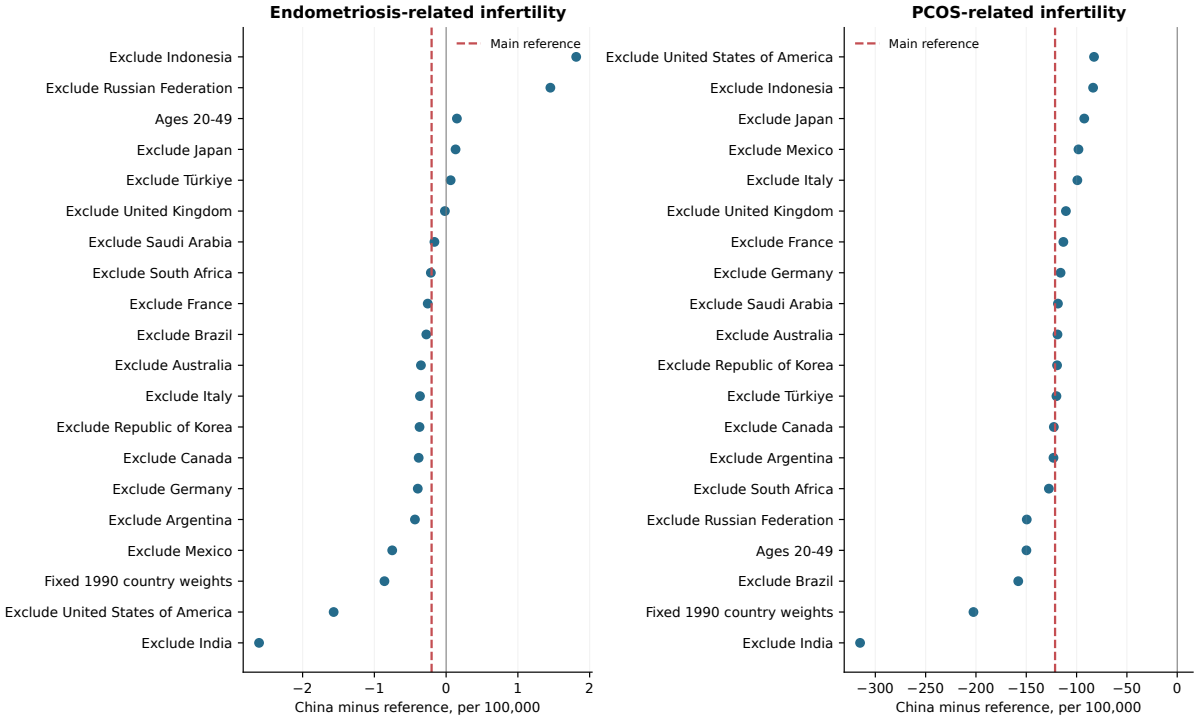


Figure S1: Sensitivity of the 2023 prevalence difference to reference-country exclusions, fixed 1990 country shares within ages, and restriction to ages 20–49. Dashed lines show the main 15–49 comparison. PCOS signs are stable; endometriosis signs vary. All are point estimates.

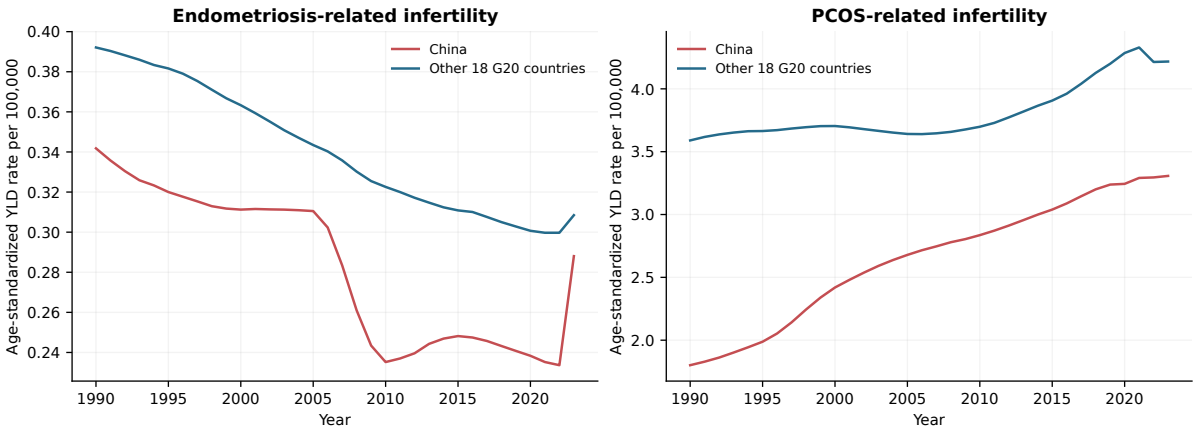


Figure S2: Restricted-age standardized YLD rates for both infertility causes, 1990–2023. These are infertility-impairment YLD rates rather than the total disability burden of either underlying disease.

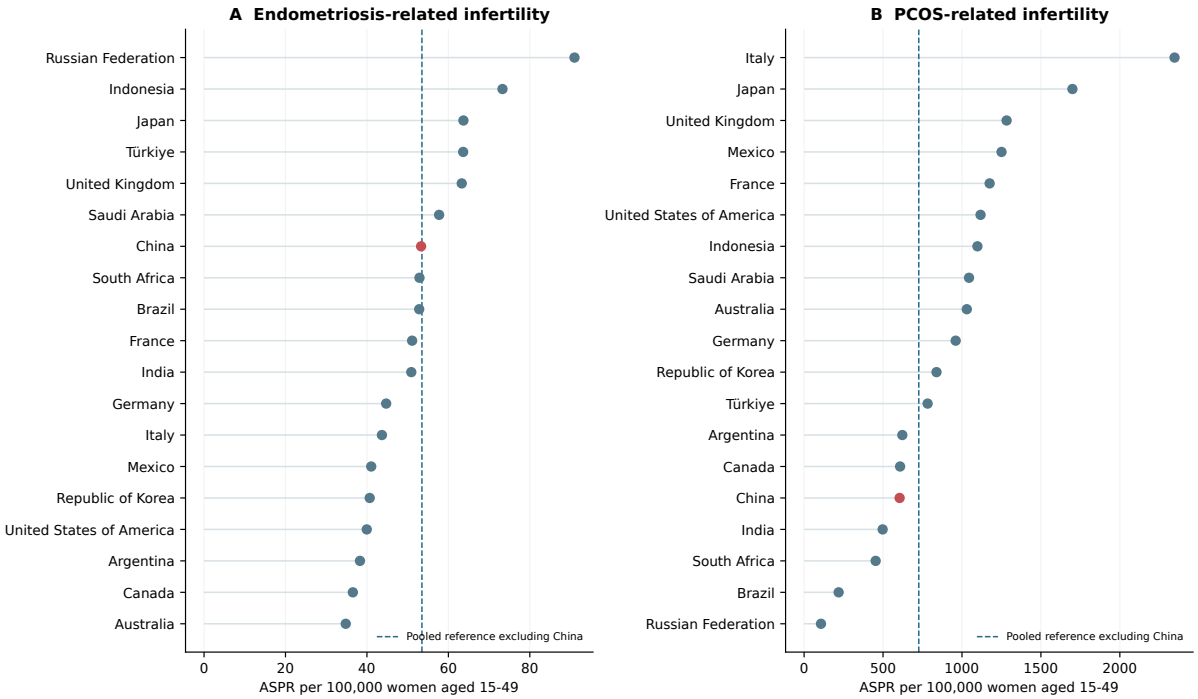


Figure S3: Country-specific prevalence rates in 2023, restricted to ages 15–49. Dashed lines are the pooled reference. Descriptive point estimates only.

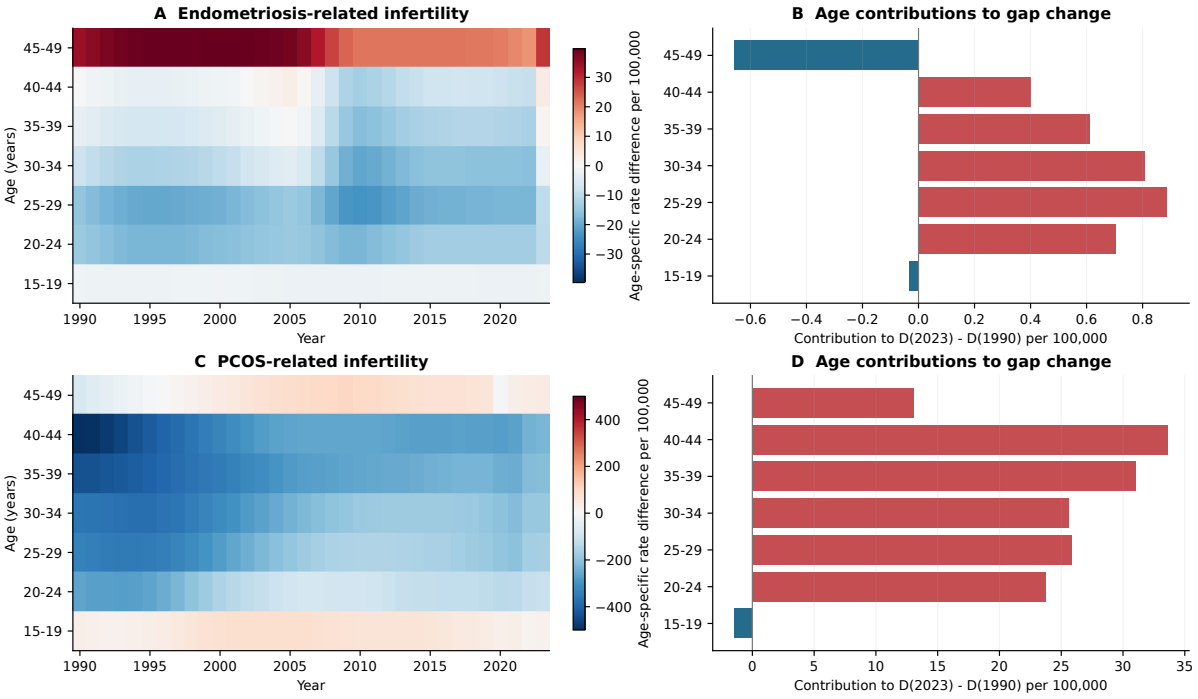


Figure S4: Age contrasts and standard-weighted contributions to changes in China minus reference. Contributions are not case counts or causal effects.

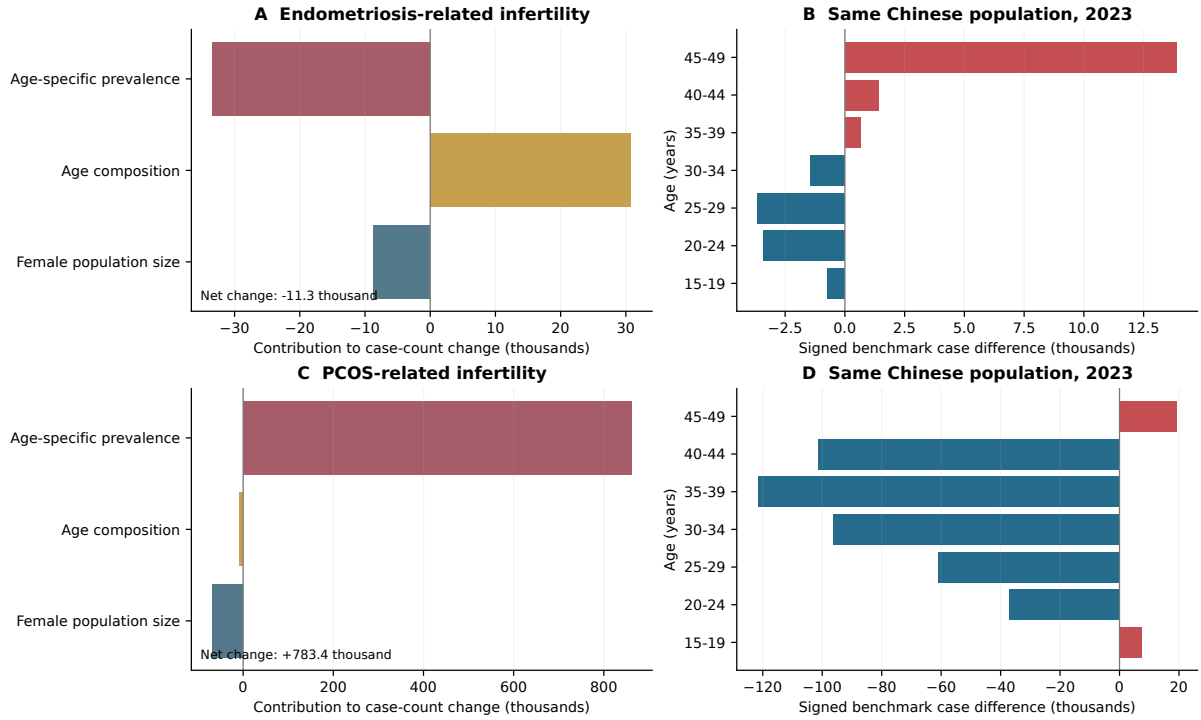


Figure S5: China's case-count decomposition and same-population benchmarks. The latter are signed arithmetic contrasts using China's actual age distribution, not preventable cases.

1.5 Exploratory NSFG age-standardization and co-diagnosis sensitivity

This post hoc analysis used the original CDC female respondent PUF ZIP and expanded CSV; both SHA-256 checks matched the archived research-source manifest (full digests in the machine-readable audit). There were 5,586 original respondents; 4,855 were aged 20–49 at interview. The one additional respondent aged 50 at interview but aged 49 at screening is included in the official 4,856-person age-domain tabulation; its inclusion did not materially alter the earlier CDC replication results. Diagnosis responses of 8, 9 and missing were excluded rather than treated as no diagnosis. Surgical sterility (FECUND=1–2) was not counted as impaired fecundity.

For diagnosis group g and age group a, let $\widehat{p}_{ga} = \widehat{Y}_{ga}/\widehat{N}_{ga}$ be the survey-weighted broad current fertility problem prevalence, with $\widehat{Y}_{ga}$ the weighted outcome numerator. Let $q_a = \widehat{N}_a / \sum_b \widehat{N}_b$ be the estimated survey-weighted population age distribution among all 20–49-year-olds. The standardized rate is $\widehat{R}_g = \sum_a q_a \widehat{p}_{ga}$. First-order variance accounted for variation in both $\widehat{p}_{ga}$ and the common estimated q_a; centered PSU totals were combined within all 20 strata (four clusters each) for survey-design confidence intervals using 60 degrees of freedom. The log prevalence-ratio contrast used the joint linearized covariance. Detailed reproducing code and derived data (without individual respondents) accompany this supplemental analysis.

2 Independent NSFG outcome and denominator audit

The original female respondent file contained 5,586 records; the strict interview-age 20–49 domain contained 4,855. Including one interview-age-50 respondent eligible at screening

Table S7: NSFG 2022–2023: broad fertility-problem differences under age standardization and co-diagnosis exclusions.

Diagnosis	Definition	Difference, pp (95% CI)	Ratio (95% CI)
PCOS	Crude	28.09 (22.11, 34.07)	2.65 (2.30, 3.04)
PCOS	Age std.	28.57 (22.50, 34.65)	2.67 (2.32, 3.07)
PCOS	Crude, no other diagnosis	28.32 (21.85, 34.80)	2.72 (2.33, 3.17)
PCOS	Age std., no other diagnosis	28.34 (21.95, 34.73)	2.71 (2.32, 3.15)
ENDO	Crude	14.41 (6.80, 22.02)	1.78 (1.40, 2.25)
ENDO	Age std.	18.68 (10.38, 26.98)	2.00 (1.59, 2.52)
ENDO	Crude, no other diagnosis	11.49 (4.16, 18.82)	1.70 (1.29, 2.23)
ENDO	Age std., no other diagnosis	15.89 (7.41, 24.36)	1.96 (1.49, 2.57)

Survey design-based, unadjusted or directly age-standardized within US women 20–49; all exploratory post hoc comparisons with no multiplicity correction. "No other diagnosis" removes respondents reporting the alternate condition. Intervals do not represent GBD posterior uncertainty.

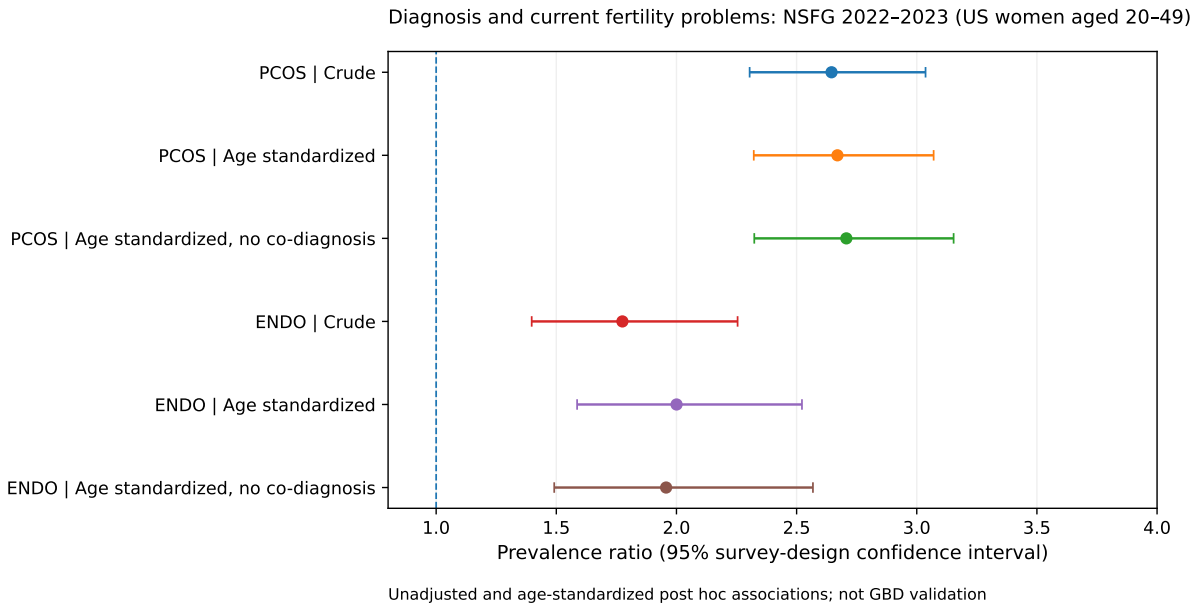


Figure S6: Sensitivity of diagnosis and broad fertility-problem prevalence ratios to age standardization and removal of co-diagnosis. The vertical dashed line is the null ratio of 1; 95% pointwise intervals use the NSFG survey design. This is a US survey association study, not validation of GBD estimates.

gave 4,856. Valid diagnosis groups were analysed separately, excluding unknown responses. Broad fertility problems were FECUND in 3–5 or INFERT=2; surgical sterility was excluded from impaired fecundity. Current union was RMARITAL in 1–2. The narrow outcome was INFERT=2, retaining valid INFERT categories 1–3 in the denominator. Narrow diagnosed numerator/domain counts were 15/173 for endometriosis and 29/252 for PCOS.

For ratio estimate $p_j = \sum_i w_i n_{ij} / \sum_i w_i d_{ij}$ the linearized respondent vector is $u_{ij} = w_i(n_{ij} - p_j d_{ij}) / \sum_i w_i d_{ij}$. Sum vectors by PSU within stratum and compute

$$\widehat{V} = \sum_h \frac{m_h}{m_h - 1} \sum_b (\boldsymbol{u}_{hb} - \overline{\boldsymbol{u}}_h)(\boldsymbol{u}_{hb} - \overline{\boldsymbol{u}}_h)^T.$$

All 80 PSUs in 20 strata were retained. The 36-by-36 matrix preserves dependence across outcome definitions. R survey and Python reconstructions agreed within approximately 10^{-15} for estimates/standard errors and 10^{-18} for covariance. These are computational checks, not GBD validation. Pointwise t_{60} intervals do not establish familywise significance or causal effects.

PCOS prevalence-difference shifts were 5.43 percentage points (95% CI 0.33–10.53) for restricting the broad-outcome population to unions, and 17.15 (10.24–24.06) for changing broad to 12-month outcome within unions. The total shift was 22.58 (16.07–29.08), using joint covariance. Endometriosis shifts were 8.28 (3.01–13.56), 7.34 (−1.44 to 16.11) and 15.62 (7.32–23.92). All are exploratory and unadjusted.

Ever fertility services were HLPPRG=1 or HLPMC=1, with definite negatives requiring both responses=5. Complete-case estimates were 30.84/12.72% for endometriosis diagnosed/not diagnosed and 34.49/11.80% for PCOS. They did not all equal the CDC rounded percentages 30.7/12.7 and 34.5/11.7. The report does not provide the full missing-response implementation; the discrepancy is not hidden or forced to agree. Six core published percentages and two diagnosis standard errors did agree to published precision.

3 Annual-chained decomposition sensitivity

The endpoint product decomposition can be applied to each adjacent year and summed. The PCOS reference-share contribution becomes +87.11 per 100,000 (57.5%), compared with endpoint +88.65 (58.5%). Both sum with their rate components to +151.54. The Chinese-rate component remains +271.51; the chained reference-rate component is −207.08. The decomposition describes time-path allocation, not a new method or GBD uncertainty.

4 GBD joint uncertainty remains unavailable

The IHME request received an automatic acknowledgement, not data or approval. Original marginal intervals do not identify cross-cell dependence. NSFG covariance cannot replace jointly indexed GBD draws. GBD derived results remain point estimates. The survey is not a proven held-out source relative to GBD and does not validate the Chinese endometriosis terminal-year change.

4.1 Supplementary GBD terminal-year source concordance and competing rate definitions

We reproduced the endometriosis and PCOS rates across all 19 study countries and 34 GBD 2023 years from the original GBD release. This is a check of the *same* model output, not an independent epidemiological replication. National annual rate-change comparisons use year-over-year percentages within the 2023 release. The selection of 2023 for extra scrutiny was post hoc; no change-point significance or pandemic effect is inferred.

Table S8: Reconciliation of Chinese GBD 2023 endometriosis-related infertility age-standardized rate conventions, per 100,000 women.

Year	Modelled cases	Restricted 15–49	Raw partial weights	Liu et al. 2026
1990	192,845	63.697	31.524	31.52
2023	181,587	53.304	26.380	26.38

The sum of published GBD 2023 weights for ages 15–49 is 0.494902. The third column renormalizes to unity within those ages; the fourth uses the original seven weights without renormalization and numerically replicates the reported rate scale. The exact cases and point estimate definitions were inspected from original GBD 2023 CSV.

Table S9: Selected country endometriosis-attributed infertility year-on-year changes in the 2023 GBD release.

Country	2022–2023 ASPR change (%)	1991–2022 max (%)
China	+23.58	+1.56
United States	+14.82	+3.88
Japan	+11.93	+1.38
Republic of Korea	−0.41	+1.56

Both percentages are from modeled GBD 2023 age-standardized rates restricted to 15–49, not independent survey measurements or posterior draws. Historical maxima are for descriptive model comparison only.

4.2 Independent population source and alternative country-share policies

The UN WPP 2024 original source is `WPP2024_PopulationByAge5GroupSex_Medium.csv.gz`, downloaded from the official UN Population Division dataset catalogue and filtered to 19 sovereign G20 countries, years 1990–2023, females ages 15–49 in five-year groups (UN Population Division, 2024). Source values in thousands of persons were multiplied by 1000. The SHA-256 values for the unmodified official compressed table and for the 4,522-row selected CSV are preserved verbatim in the machine-readable source manifest and submission archive. These are demographic data only and provide no independent estimates of infertility.

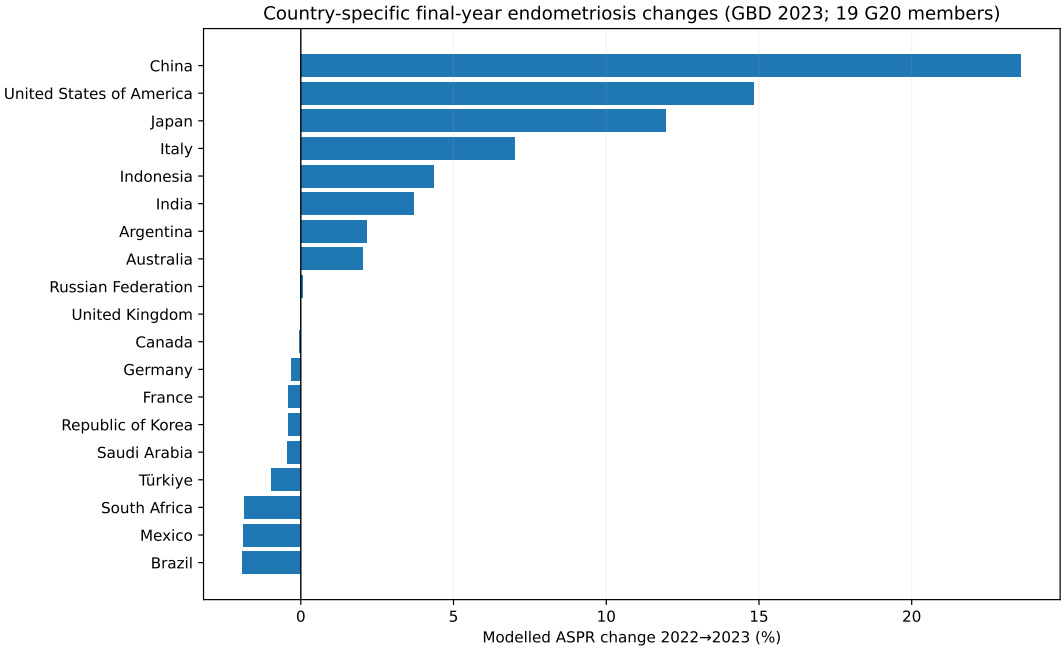


Figure S7: GBD 2023 endpoint model changes in endometriosis-attributed infertility rates across all 19 G20 countries, 2022–2023. Positive and negative bars are changes in modeled 15–49 ASPR, not evidence of population incidence changes.

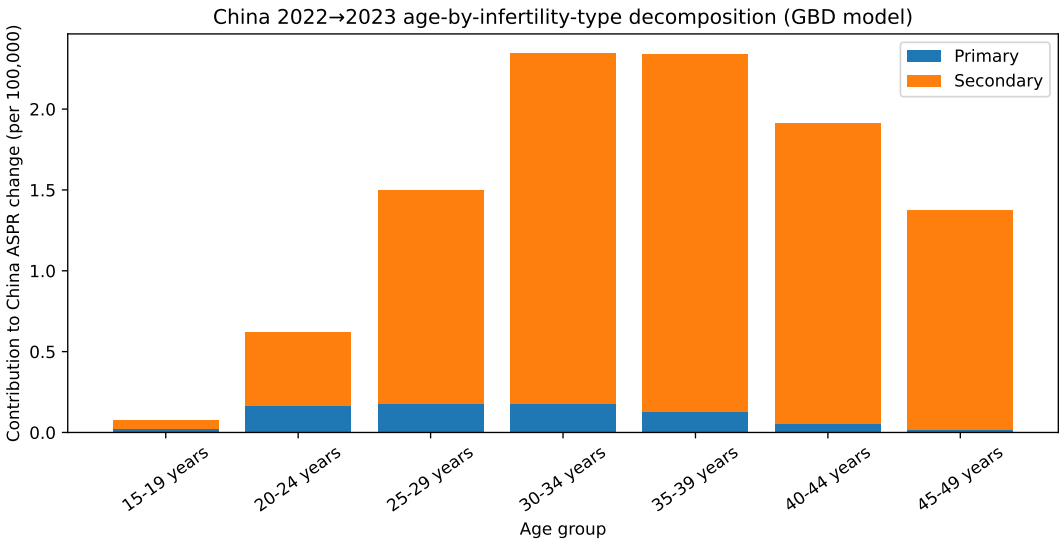


Figure S8: Standard-weighted five-year-age and infertility-type contributions to the modeled Chinese endometriosis-attributed infertility 2022–2023 rate change. Sum of primary and secondary contributions equals the total 15–49 ASPR change (10.1723 per 100,000).

Table S10: PCOS-related secondary infertility: 2023 reference sensitivity to country-population weighting.

Population source	Country-share policy	Reference rate	China–reference gap
GBD 2023	Contemporary 2023 shares	505.56	+2.16
UN WPP 2024	Contemporary 2023 shares	505.46	+2.26
UN WPP 2024	Fixed 1990 shares	556.06	−48.34
Equal countries	Each of 18 countries weighted 1/18	635.61	−127.89

All values per 100,000 women aged 15–49, with the identical underlying GBD 2023 age-specific rates. Fixed shares are within five-year age groups; equal-country means equal weighting inside each age group. Population-source sensitivity does not estimate sampling or posterior uncertainty.

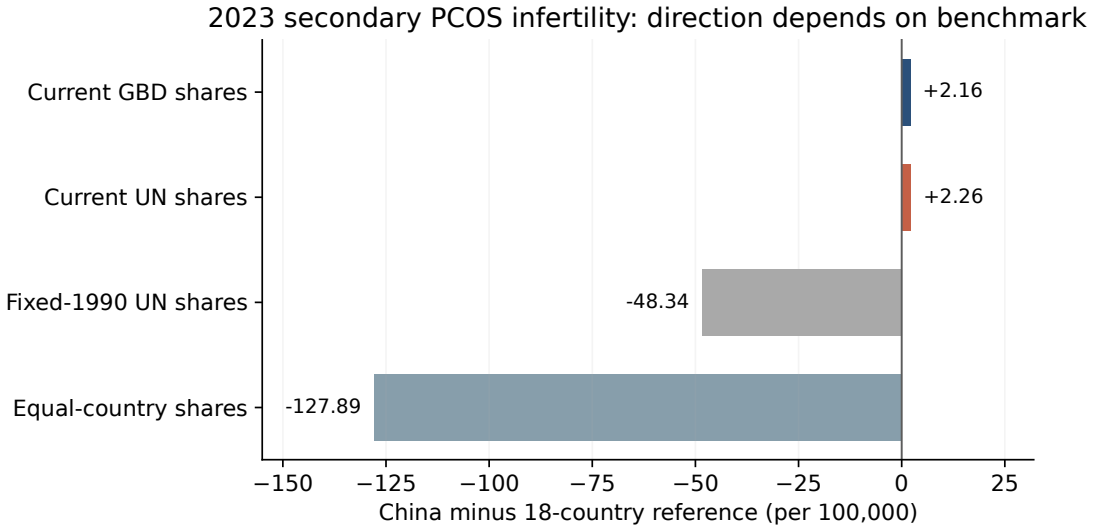


Figure S9: Benchmark choice reverses the sign of the near-zero PCOS secondary infertility difference in 2023. Different benchmark definitions are different estimands, not confidence intervals.

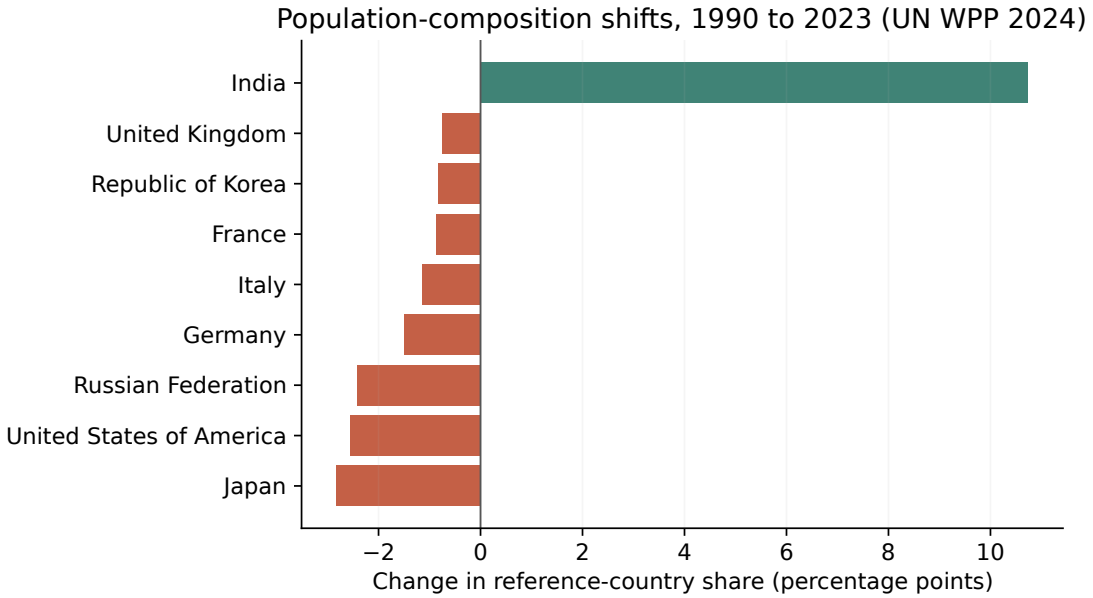


Figure S10: Largest changes in age-weighted reference-country population shares from 1990 to 2023 using WPP 2024. China is excluded from the reference.